

HW 1

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9/29/2026

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- (a) Given random variable X where $a \leq X \leq b$, we can prove that $X - E[X]$ is $\frac{(b-a)^2}{4}$ -subgaussian. Let $Y = X - E[X]$ and the log MGF is $\psi(\lambda) = \log E[e^{\lambda Y}]$. Using the identity from the hint, we can bound $\psi''(\lambda)$ as follows:

$$\begin{aligned}\text{Let } M(\lambda) &= E[e^{\lambda Y}] \\ \psi(\lambda) &= \log M(\lambda) \\ \psi'(\lambda) &= \frac{M'(\lambda)}{M(\lambda)} = \frac{E[Y e^{\lambda Y}]}{E[e^{\lambda Y}]} \\ \psi''(\lambda) &= \frac{M''(\lambda)M(\lambda) - (M'(\lambda))^2}{(M(\lambda))^2} = \frac{E[Y^2 e^{\lambda Y}]E[e^{\lambda Y}] - (E[Y e^{\lambda Y}])^2}{(E[e^{\lambda Y}])^2}\end{aligned}$$

Since $Y = X - E[X]$ and subtracting a constant does not change variance:

$$\psi''(\lambda) = \frac{E[Y^2 e^{\lambda Y}]}{E[e^{\lambda Y}]} - \left(\frac{E[Y e^{\lambda Y}]}{E[e^{\lambda Y}]} \right)^2 = \text{Var}(\tilde{X}) \leq \frac{(b-a)^2}{4}$$

where $\tilde{X}$ is a random variable that satisfies $a \leq \tilde{X} \leq b$. Therefore, we have:

$$\psi(\lambda) \leq \int_0^\lambda \int_0^t \frac{(b-a)^2}{4} ds dt = \frac{(b-a)^2 \lambda^2}{8}$$

A mean zero random variable is σ^2 -subgaussian if $\psi(\lambda) \leq \frac{\sigma^2 \lambda^2}{2}$. Thus, we have:

$$\begin{aligned}\frac{\sigma^2 \lambda^2}{2} &= \frac{(b-a)^2 \lambda^2}{8} \\ \sigma^2 &= \frac{(b-a)^2}{4}\end{aligned}$$

Thus, $X - E[X]$ is $\frac{(b-a)^2}{4}$ -subgaussian.

- (b) The constant $\frac{(b-a)^2}{4}$ is the best possible constant. To show this, we use the example of a random variable X that takes on the values a and b with equal probability. Then, we have:

$$E[X] = \frac{a+b}{2}$$

$$Var(X) = E[X^2] - (E[X])^2 = \frac{a^2+b^2}{2} - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{4}$$

Let $Y = X - E[X]$. Then, we have:

$$E[Y] = E[X - E[X]] = E[X] - E[X] = 0$$

$$Var(Y) = Var(X) = \frac{(b-a)^2}{4}$$

Now, suppose Y is σ^2 -subgaussian. Then

$$\psi_Y(\lambda) = \log E[e^{\lambda Y}] \leq \frac{\sigma^2 \lambda^2}{2}$$

Since $\psi_Y''(0) = Var(Y)$ and $\left(\frac{\sigma^2 \lambda^2}{2}\right)'' = \sigma^2$, the subgaussian bound near $\lambda = 0$ implies

$$\frac{(b-a)^2}{4} \leq \sigma^2$$

Thus, $Y = X - E[X]$ cannot be σ^2 -subgaussian for any $\sigma^2 < \frac{(b-a)^2}{4}$. Therefore, the constant $\frac{(b-a)^2}{4}$ cannot be improved in general.

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Let X be a random variable where $E[X] = 0$, $Var(X) \leq \sigma^2$, and $|X| \leq b$ almost surely for a finite constant b . For any $\lambda \in (-3/b, 3/b)$, we can show that $\psi(\lambda) \leq \frac{\lambda^2 \sigma^2}{2(1-|\lambda|b/3)}$ by using the Taylor expansion of $e^{\lambda X}$:

$$e^{\lambda X} = 1 + \lambda X + \frac{\lambda^2 X^2}{2!} + \frac{\lambda^3 X^3}{3!} + \dots$$

$$E[e^{\lambda X}] = 1 + \frac{\lambda^2 E[X^2]}{2!} + \frac{\lambda^3 E[X^3]}{3!} + \dots$$

We can define an upper bound $\lambda^n X^n \leq |\lambda|^n |X|^n$, but since $|X| \leq b$, we have $|X|^n \leq b^{n-2} |X|^2$. So:

$$\begin{aligned} E[e^{\lambda X}] &\leq 1 + E[X^2] \left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \dots \right) \\ E[X] &= 0, \quad E[X^2] = \text{Var}(X) \leq \sigma^2 \therefore \\ E[e^{\lambda X}] &\leq 1 + \sigma^2 \left(\frac{\lambda^2}{2!} + \frac{|\lambda|^3 b}{3!} + \frac{|\lambda|^4 b^2}{4!} + \dots \right) \\ E[e^{\lambda X}] &\leq 1 + \sigma^2 \left(\frac{\lambda^2}{2} \right) \left(1 + \frac{|\lambda|b}{3} + \frac{(|\lambda|b)^2}{12} + \dots \right) \end{aligned}$$

But we can recognize the series as a geometric series for the third term onward since $\frac{|\lambda|^4 b^2}{4!} \leq \frac{\lambda^2 b^2}{2} \left(\frac{|\lambda|b}{3} \right)^2$. Thus, we can write:

$$E[e^{\lambda X}] \leq 1 + \frac{\lambda^2 \sigma^2}{2} \left(1 + \frac{|\lambda|b}{3} + \left(\frac{|\lambda|b}{3} \right)^2 + \dots \right)$$

We know that $|\lambda| < \frac{3}{b}$, so $\frac{|\lambda|b}{3} < 1$ and we can sum the geometric series using $r = \frac{|\lambda|b}{3}$:

$$\begin{aligned} E[e^{\lambda X}] &\leq 1 + \frac{\lambda^2 \sigma^2}{2} \left(\frac{1}{1 - |\lambda|b/3} \right) \\ E[e^{\lambda X}] &\leq 1 + \frac{\lambda^2 \sigma^2}{2(1 - |\lambda|b/3)} \\ \log(1+x) &\leq x \therefore \\ \log E[e^{\lambda X}] = \psi(\lambda) &\leq \frac{\lambda^2 \sigma^2}{2(1 - |\lambda|b/3)} \end{aligned}$$